

The Brazilian eCommerce

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Executive Summary

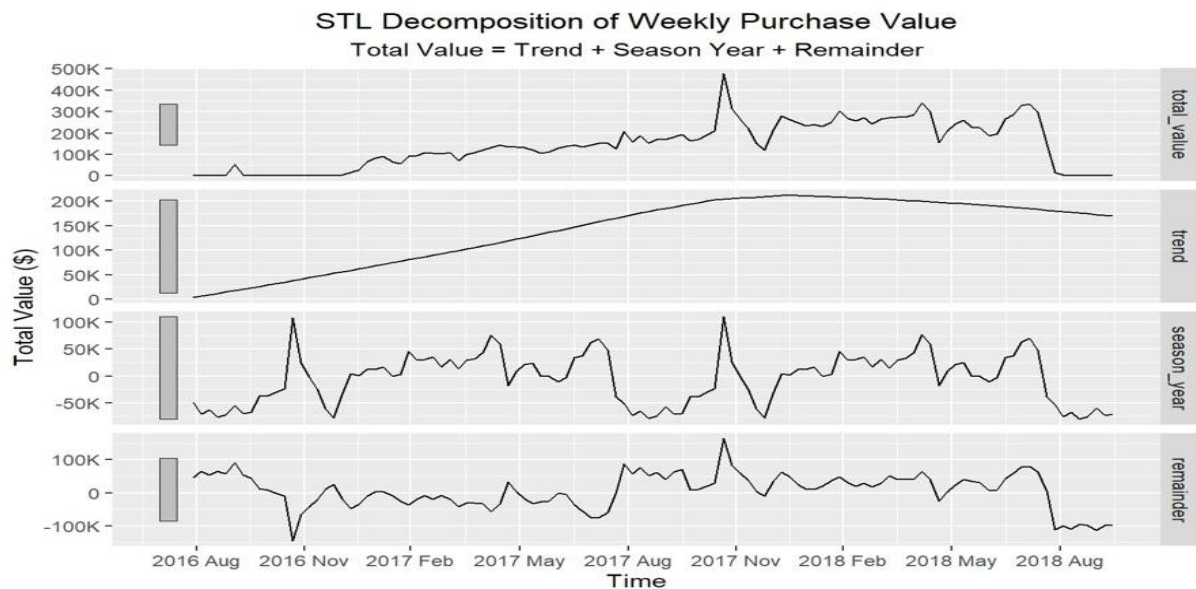
Olist connects small and medium-sized businesses with major online retail channels in Brazil. Managing this complex network requires careful planning, smooth logistics, and smart inventory control.

This report analyzes Olist's past sales data (olist.sqlite) by focusing on two main areas:

1. **Exploratory Data Analysis (EDA):** Looking at overall sales trends and revenue by category to find weaknesses in the platform.
2. **Predictive Modelling Framework:** Building a forecasting tool (using R-tidymodels) to predict next week's revenue. This helps Olist and its sellers improve planning, efficiency, and profits.

Goal 1: Exploratory Data Analysis and Strategic Findings

Finding 1: Olist's purchase value grew rapidly over time, while seasonality was less clear



The first major finding is that Olist's weekly purchase value increased substantially over the period under observation, suggesting strong platform growth. Although weekly purchase value fluctuated over time, the STL decomposition shows that the trend component was the dominant pattern in the data. The trend increased steadily through 2017 and reached its highest level around early 2018 before flattening near the end of the dataset.

To investigate this, we combined the order and payment data by purchase week and calculated the total weekly purchase value. We then applied STL decomposition to separate the time series into three

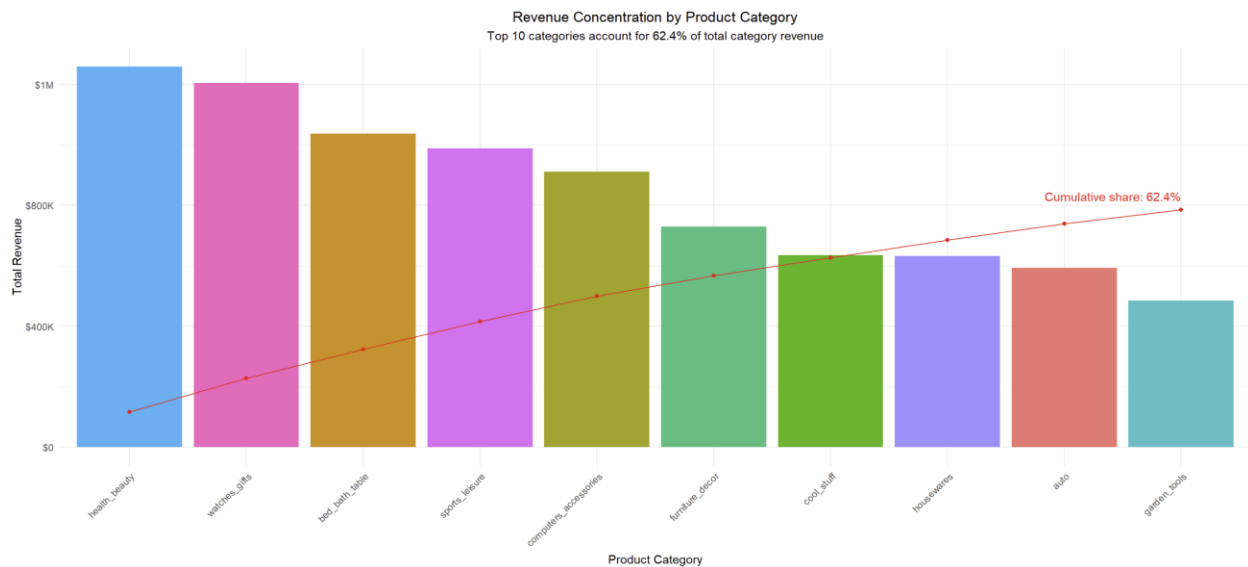
components: trend, seasonal variation, and remainder. Breaking the STL into components helped distinguish long-term growth from repeated seasonal movement and short-term noise.

The decomposition suggests that Olist's purchase value was more influenced by long-term market growth than by a clear seasonal cycle. The trend panel shows a strong upward movement across most of the period, while the seasonal component is smaller relative to the overall increase in purchase value and shows no clear pattern. Therefore, the main business insight is that Olist was not simply experiencing normal seasonal ups and downs; it was operating in a period of rapid expansion.

The sharp decline at the end of the graph should be interpreted carefully. Since the available dataset ends on July 5, 2018, the final weeks likely contain incomplete transaction data. As a result, the drop near the end of the series is more likely due to the data cutoff than to a real collapse in customer demand.

This finding matters because rapid growth creates operational challenges. If Olist treats every increase or decrease in sales as a seasonal pattern, management may miss the larger issue: the platform needs enough operational capacity to support sustained growth. Olist should therefore focus on scalable improvements such as seller onboarding, payment processing, customer support, and delivery coordination. However, because operational resources are limited, Olist should not treat all product categories equally. The next step is to identify which categories generate the most revenue so that growth planning can focus on the areas with the greatest financial impact.

Finding 2: Revenue is concentrated in a small number of product categories



The second major finding is that Olist's revenue is highly concentrated among a small number of product categories. Although the dataset contains 72 product categories, the top 10 categories account for 62.4% of total category revenue. This product concentration shows that Olist's marketplace revenue is not evenly distributed across categories. Instead, a relatively small group of categories drives most of the platform's sales.

The chart also shows a clear separation among the highest-revenue categories. Health/beauty, watches/gifts, bed/bath/table, sports/leisure, and computer accessories are the top five revenue-generating categories, and they are the only categories that exceed roughly \$800,000 in total revenue. The higher revenue concentration makes the top five categories especially important from a business perspective because changes in their demand or delivery performance could have a larger effect on total revenue than changes in lower-revenue categories.

To discover this finding, we joined the product, order item, and product category translation tables. Joining the tables allowed us to connect each sold item to an English product category name and calculate total revenue by category using item prices. We then ranked product categories by revenue from highest to lowest and calculated the cumulative revenue share to measure the extent to which total revenue was explained by the largest categories.

Ranking categories by revenue and computing cumulative revenue are important because revenue concentration creates both opportunities and risks. On the opportunity side, Olist can prioritize high-revenue categories when planning seller support, promotions, inventory preparation, and delivery coordination. Improving performance in these categories is likely to have a meaningful impact on overall revenue. However, concentration also increases risk. If one of the major categories experiences stock shortages, delayed deliveries, or lower customer satisfaction, Olist's overall revenue performance could be affected.

This finding also supports the prediction objective in Goal 2. Since category-level revenue is highly uneven, forecasting next week's revenue by product category can help Olist identify which categories are likely to drive future sales and prepare sellers and logistics partners before demand changes occur.

Goal 2: Predicting Next-Week Revenue by Product Category

Business Objective

The business objective for Goal 2 is to predict next week's revenue for each product category. This prediction can help Olist improve operational efficiency by identifying which categories are expected to generate the most sales in the coming week. If Olist can anticipate category-level demand in advance, it can better coordinate with sellers, plan logistics capacity, and reduce the risk of stock shortages or delivery delays in high-revenue categories.

Next week's prediction objective connects directly to the revenue concentration finding from Goal 1. Since a relatively small number of product categories account for a large share of total revenue, forecasting category-level revenue can help Olist focus on the categories most likely to affect overall marketplace performance.

Data Preparation and Feature Engineering

To build the prediction dataset, we grouped transactions by product category and purchase week. The target variable was `future_revenue`, representing next week's revenue for the same product category. The

future_revenue variable was created using a lead function, so each row uses current and past information to predict next week's revenue.

Several predictors were engineered to capture category-level demand and operational conditions. These included current weekly revenue, previous week revenue, the number of active sellers, average review score, late-delivery proportion, four-week rolling average revenue, and four-week rolling revenue volatility. These prediction variables were selected because they describe both demand momentum and operational performance.

We also cleaned missing and inconsistent values before modelling. Eight orders were identified as delivered, but did not have a recorded customer delivery date. Since delivery timeliness could not be determined for these observations, they were excluded from the late-delivery analysis. Missing weekly review scores were imputed using the average review score within the corresponding product category. For category-weeks with no observed delivery records, the late-delivery proportion was set to zero. These decisions allowed us to keep the weekly category-level dataset complete while avoiding misleading late-delivery calculations.

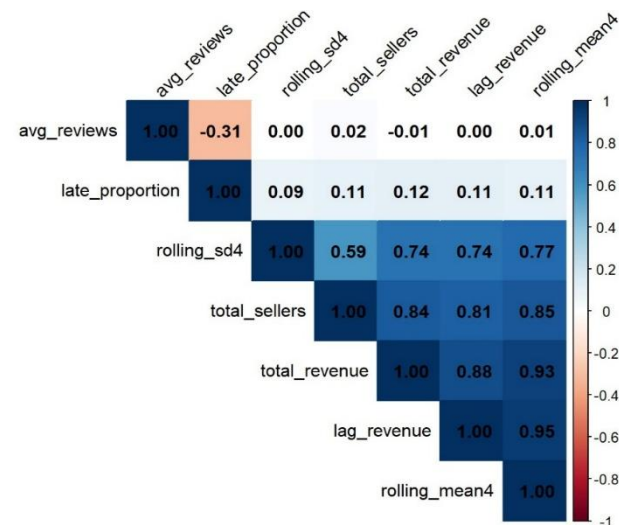
A time-based split was used instead of a random split. The time-based approach means the data were kept in chronological order, with older observations used for training and newer observations for testing. The first 70% of the prepared dataset was used for training, and the remaining 30% was reserved for testing. Because the prepared dataset contained fewer than 10,000 observations, the 70:30 split provided enough data for model training while still leaving a meaningful test set for evaluating forecasting performance.

Models Tested

We compared several model types to determine which approach best predicted next-week category revenue. For each model type, we tested different predictor combinations and, when applicable, adjusted model parameters to improve forecasting performance.

First, we tested linear regression models using different combinations of predictors. Linear regression was useful as a baseline model because it is easy to interpret and indicates whether revenue can be predicted by a mostly linear relationship.

Second, we tested PCA-based linear regression. The correlation matrix and VIF results indicated that some predictors were strongly correlated, particularly revenue-based variables such as current revenue, lagged revenue, and rolling average revenue. PCA was used to reduce multicollinearity by transforming the original predictors into principal components. Breaking the prediction variables into principal components enabled us to test whether a lower-dimensional representation of the data could improve predictive performance.



Third, we tested regression tree models. Regression trees are useful because they can capture non-linear relationships and interactions between predictors. Unlike linear regression, they do not require the relationship between predictors and revenue to be strictly linear. The regression tree models were tuned by changing parameters such as tree depth and the minimum number of observations required in a terminal node.

Finally, we tested random forest models. Random forests improve on single decision trees by building many trees and averaging their predictions. The averaging method generally reduces overfitting and produces more stable predictions. The random forest models were tuned by changing the number of trees, the number of predictors considered at each split, and the minimum number of observations required in a terminal node.

VIF Table	
Predictor	VIF
total_revenue	7.797976
lag_revenue	9.470140
total_sellers	4.087960
avg_reviews	1.112398
late_proportion	1.126171
rolling_sd4	2.645790
rolling_mean4	16.702361

Model Evaluation and Selection

Final Model Comparison				
Performance on the time-based testing set				
Model	Best Version	MAE	RMSE	R ²
Linear Regression	total_revenue + lag_revenue	1,121.39	2,137.53	80.31%
PCA Linear Regression	7 principal components	1,086.98	2,161.45	79.82%
Regression Tree	tree_depth = 5, min_n = 5, cost_complexity = 0.001	1,168.33	2,128.57	80.46%
Random Forest	500 trees, mtry = 1, min_n = 20	1,122.26	2,099.07	80.98%

The models were evaluated on the testing set using RMSE, MAE, and R-squared. RMSE and MAE measure prediction error, while R-squared measures how much variation in future revenue the model explains on unseen data.

The random forest model achieved the best overall test performance. It achieved the highest testing R-squared and a lower prediction error than the other model types. The high R-squared and low prediction error suggest that the random forest better captured the non-linear relationships and interactions among the predictors, including the combined effect of current revenue, lagged revenue, rolling-average revenue, and delivery performance on next-week revenue.

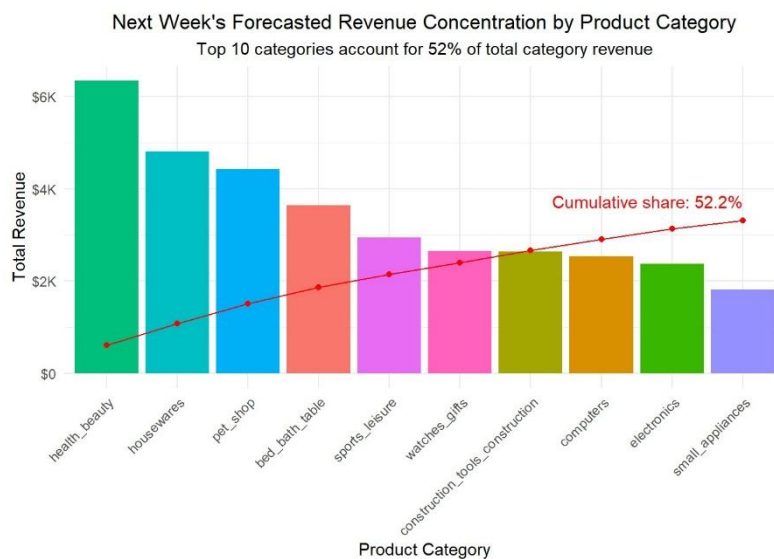
Based on the final comparison, the selected model was the best-performing random forest model. This model was then used to generate the next-week revenue forecast for each product category.

Final Business Insight & How Olist Can Use This Report

The final random forest model was used to forecast next-week revenue by product category. The forecasted results showed which categories are expected to generate the highest revenue in the next period. The forecasted revenue gives Olist a practical tool for short-term operational planning.

From a business perspective, the model can support three main decisions. First, Olist can use the forecast to identify high-demand categories before the demand occurs. Knowing high-demand categories in advance allows the company to communicate with sellers and encourage them to prepare sufficient inventory. Second, Olist can use the forecast to coordinate delivery capacity for categories expected to generate higher order volume. Third, management can use the forecast to monitor whether the same high-revenue categories from Goal 1 are expected to continue driving revenue in the near future.

The model should not be interpreted as a perfect prediction system. Instead, it should be used as an early warning tool. If the model predicts that a category's revenue will increase next week, Olist can investigate whether sellers and logistics partners are prepared. If the model predicts lower revenue, Olist can avoid over-allocating resources to that category. Therefore, predicting next week's revenue for each category helps improve efficiency while still supporting customer satisfaction.



Top 10 Predicted Product Categories by Next Week Revenue			
Product Category	Latest Available Date	Next Forecasted Date	Forecasted Revenue
health_beauty	2018 W35	2018 W36	\$6,338.31
housewares	2018 W35	2018 W36	\$4,803.04
pet_shop	2018 W34	2018 W35	\$4,429.51
bed_bath_table	2018 W35	2018 W36	\$3,642.31
sports_leisure	2018 W35	2018 W36	\$2,939.67
watches_gifts	2018 W35	2018 W36	\$2,648.97
construction_tools_construction	2018 W35	2018 W36	\$2,642.11
computers	2018 W34	2018 W35	\$2,532.92
electronics	2018 W34	2018 W35	\$2,369.00
small_appliances	2018 W34	2018 W35	\$1,818.22

The next week's forecast suggests that health/beauty will remain the highest-revenue category, followed by housewares, pet shop, bed/bath/table, and sport/leisure. Comparing it to the high-revenue categories identified in the historical analysis, we can see that many of the same categories remain important in the near-term forecast. However, the appearance of categories such as housewares and pet shop suggests that short-term demand can differ from the overall historical ranking.

A limitation of this model is that it uses historical category revenue and operational variables. However, it does not include external factors such as marketing campaigns, holidays, economic market conditions, or competitor actions. These external missing factors may explain some of the remaining prediction error.

Usage of AI

For this report, we used AI to correct our grammar, generate ideas, and debug code. For instance, we were having issues with NAs and data gaps, so we asked AI and Erfan for help overcoming them. Additionally, we were unsure how to import SQL data into R, so we used AI to help us. AI is useful for

generating good ideas for solutions, but because the data is large and complex, it sometimes makes mistakes. However, with enough effort and Erfan's advice, we were able to figure it out.

Appendix

Data Preparation

The Prediction Category Table										
First 10 Observations										
Product Category	Current Date	Predicted Date	Future Revenue	Revenue	Lag Revenue	Total Sellers	Avg.Review	Late Proportion	4 Week Rolling SD	4 Week Rolling Avg.
agro_industry_and_commerce	2017 W05	2017 W06	114.89	43.98	43.98	1	4.000000	0	0.00000	43.98000
agro_industry_and_commerce	2017 W06	2017 W07	65.97	114.89	43.98	2	3.500000	0	40.93991	67.61667
agro_industry_and_commerce	2017 W07	2017 W08	21.99	65.97	114.89	1	4.000000	0	33.43743	67.20500
agro_industry_and_commerce	2017 W08	2017 W09	0.00	21.99	65.97	1	5.000000	0	35.31891	58.16200
agro_industry_and_commerce	2017 W09	2017 W10	0.00	0.00	21.99	0	4.021978	0	44.11503	49.36600
agro_industry_and_commerce	2017 W10	2017 W11	59.99	0.00	0.00	0	4.021978	0	49.51184	40.57000
agro_industry_and_commerce	2017 W11	2017 W12	22.00	59.99	0.00	1	1.000000	0	31.84557	29.59000
agro_industry_and_commerce	2017 W12	2017 W13	0.00	22.00	59.99	1	3.000000	0	24.51526	20.79600
agro_industry_and_commerce	2017 W13	2017 W14	0.00	0.00	22.00	0	4.021978	0	26.16452	16.39800
agro_industry_and_commerce	2017 W14	2017 W15	0.00	0.00	0.00	0	4.021978	0	26.16452	16.39800

Linear Regression

PCA Linear Regression Table						
Model	MAE	RMSE	R ²	Adjusted R ²	num_comp	
future_revenue ~ PC1	1122.379	2272.841	77.76%	75.78%	1	
future_revenue ~ PC1+PC2	1122.614	2272.791	77.76%	75.87%	2	
future_revenue ~ PC1+PC2+PC3	1118.808	2271.657	77.78%	75.87%	3	
future_revenue ~ PC1+PC2+PC3+PC4	1128.658	2310.095	76.94%	78.66%	4	
future_revenue ~ PC1+PC2+PC3+PC4+PC5	1156.149	2355.830	76.03%	79.37%	5	
future_revenue ~ PC1+PC2+PC3+PC4+PC5+PC6	1101.514	2210.530	78.89%	79.56%	6	
future_revenue ~ PC1+PC2+PC3+PC4+PC5+PC6+PC7	1086.977	2161.447	79.82%	79.90%	7	

PCA Model Fit Statistics

term	estimate	std.error	statistic	p.value
(Intercept)	1946.08191	29.50747	65.952177	0.000000e+00
PC1	1751.20270	14.21671	123.179209	0.000000e+00
PC2	106.92052	25.65424	4.167752	3.139235e-05
PC3	45.64298	36.22723	1.259908	2.077738e-01
PC4	1053.88465	44.51568	23.674459	2.719087e-116
PC5	953.65248	80.24879	11.883700	4.768095e-32

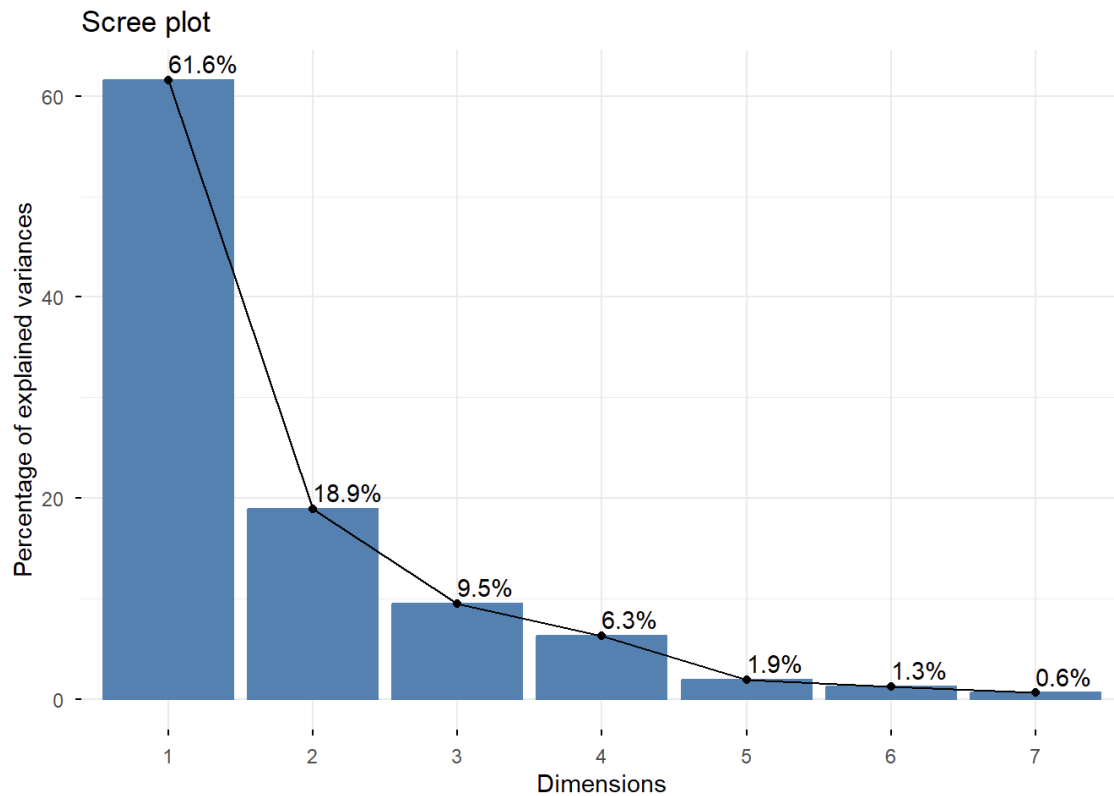
r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual	nobs
79.39%	79.37%	1896.759	3178.755	0	5	-37047.98	74109.96	74154.25	14844087050	4126	4132

.metric	.estimate
mae	1156.148860
rmse	2355.830377
rsq	0.760258

PCA VIF Proof – No Multicollinearity

PC1	PC2	PC3	PC4	PC5
1	1	1	1	1

PCA Scree Plot



Non-PCA Model Table

Non-PCA Linear Regression Table				
Model	MAE	RMSE	R ²	Adjusted R ²
future_revenue ~ total_revenue	1164.895	2173.488	79.70%	75.80%
future_revenue ~ total_revenue+lag_revenue	1121.393	2137.533	80.31%	77.00%
future_revenue ~ total_revenue+lag_revenue+total_sellers	1138.931	2307.118	77.00%	78.80%
future_revenue ~ total_revenue+lag_revenue+total_sellers+avg_reviews	1139.113	2307.200	77.00%	78.80%
future_revenue ~ total_revenue+lag_revenue+total_sellers+avg_reviews+late_proportion	1140.634	2307.224	77.00%	78.80%
future_revenue ~ total_revenue+lag_revenue+total_sellers+avg_reviews+late_proportion+rolling_sd4	1141.554	2292.898	77.29%	78.90%
future_revenue ~ total_revenue+lag_revenue+total_sellers+avg_reviews+late_proportion+rolling_sd4+rolling_mean4	1086.977	2161.447	79.82%	79.90%

Regression Tree

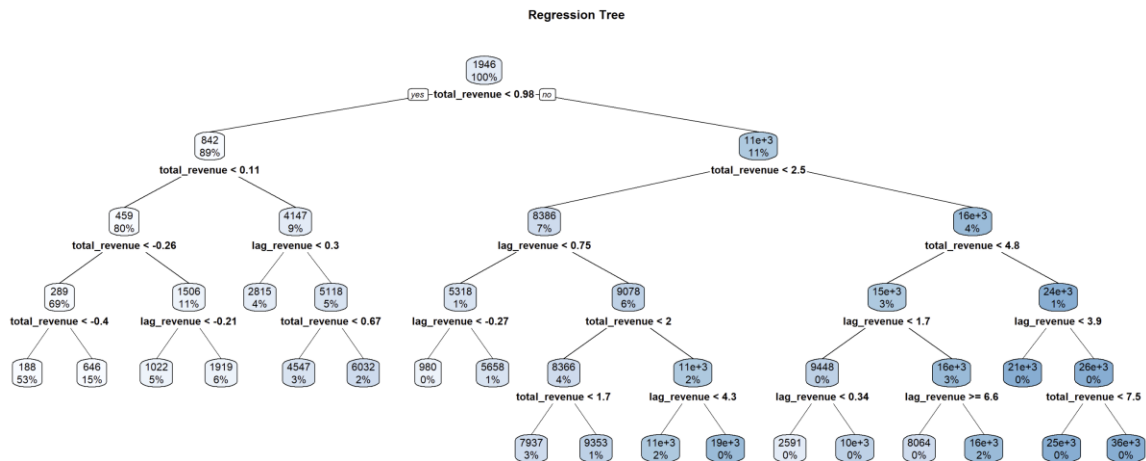
The Table

Regression Tree Table

First 10 Observations

Model	MAE	RMSE	R ²	tree_depth	min_n	cost_complexity
future_revenue ~ total_revenue+lag_revenue	1168.330	2128.571	80.46%	5	5	0.001
future_revenue ~ total_revenue+lag_revenue	1169.653	2139.168	80.29%	5	2	0.001
future_revenue ~ total_revenue+lag_revenue	1172.812	2137.296	80.27%	5	10	0.001
future_revenue ~ total_revenue+lag_revenue	1179.341	2149.693	80.26%	8	5	0.001
future_revenue ~ total_revenue+lag_revenue	1166.824	2149.739	80.06%	5	10	0.000
future_revenue ~ total_revenue+lag_revenue	1185.876	2160.127	80.05%	8	10	0.001
future_revenue ~ total_revenue+lag_revenue	1180.796	2167.900	79.99%	10	2	0.001
future_revenue ~ total_revenue+lag_revenue	1169.906	2156.860	79.94%	5	5	0.000
future_revenue ~ total_revenue+lag_revenue	1166.490	2158.125	79.94%	5	2	0.000
future_revenue ~ total_revenue+lag_revenue	1191.821	2172.874	79.87%	10	10	0.001

Tree Plot



Random Forest

Top 5 Random Forest Results							
Model	MAE	RMSE	R ²	trees	mtry	min_n	
future_revenue ~ total_revenue+lag_revenue	1122.263	2099.074	80.98%	500	1	20	
future_revenue ~ total_revenue+lag_revenue	1123.054	2105.103	80.88%	750	1	20	
future_revenue ~ total_revenue+lag_revenue	1121.983	2106.414	80.86%	1000	1	20	
future_revenue ~ total_revenue+lag_revenue	1135.355	2112.262	80.85%	500	2	20	
future_revenue ~ total_revenue+lag_revenue	1135.024	2112.825	80.83%	750	2	20	